



2 Unit Mathematics Assessment 27/2/1998

Year 11

Question 1.

- 2 a) Evaluate $\frac{4 + \sqrt{40}}{4}$ correct to 2 decimal places.
- 2 b) A person's heart beats approximately 72 times per minute. Assuming there are 365 days in every year, find how many times a heart would beat in 70 years. Give your answer in scientific notation correct to three significant figures.
- c) The radius, r , of a conical flask of height h and volume V is given by $r = \sqrt{\frac{3V}{\pi h}}$. A manufacturer is required to produce two conical flasks each having a volume of $1\,000\text{ cm}^3$. Calculate, in centimetres, and correct to one decimal place
- 1 (i) the radius of the conical flask of height 50 cm.
- 2 (ii) the radius of the conical flask with height and radius of equal magnitude.
- 2 d) Graph on the number line the solution set of $-12 \leq \frac{2x}{3} < 4$.
- 1 e) "At Gosford High there are four houses, so that the probability that O.S.U. will win the next swimming carnival is $\frac{1}{4}$ ". Is this statement true or false? Explain in no more than one sentence.
- 2 f) Express with a rational denominator $\frac{6}{7 - 2\sqrt{5}}$.

Question 2.

- 2 a) Solve for x : $|1 - 2x| = 5$
- 2 b) Find the exact values of x if $(x - 1)^2 = 7$
- 2 c) At Octopus Communications' annual sale, all mobile phones were discounted by 40%. Arun paid \$156 for a mobile phone at the sale. What was the original price of the phone?
- 2 d) If $x\sqrt{3} + 4\sqrt{y} = \sqrt{75} + \sqrt{80} + \sqrt{12}$ find the value of x and y .
- 2 e) Show that the recurring decimal $0.4\overline{7}$ is rational.

- 2 f) Solve for x : $|3x - 2| \leq 4$

Question 3.

- 2 a) Factor fully: $x^3y - xy^3$
- 2 b) Simplify: $\frac{1}{x} - \frac{1-x}{2x}$
- 1 c) Solve: (i) $4 - x = 5$
- 2 (ii) $2x^2 - 7x - 4 = 0$
- 2 (iii) $2 - (1 - x) \leq 3 + 2x$
- 2 (iv) $x^2 = 9x$
- 1 d) Find the value of x if 2^x is half of 2^{400}

Question 4.

- 2 a) Solve $|5 - 2x| \geq 3$
- 2 b) Solve $\frac{3x - 2}{4} - \frac{2x + 1}{8} = 5$
- 2 c) Factorise fully $a^2 - b^2 - 3a + 3b$
- 2 d) Factorise $8x^3 - 1\,000$
- 2 e) If $x + \frac{1}{x} = k$ what is the value of $x^2 + \frac{1}{x^2}$
- 2 f) Solve the simultaneous equations $3x - 2y = 8$ and $5x - 6y = 16$

Question 5.

- 2 a) Solve for x : $|x + 7| = 3x - 1$
- 2 b) Find in terms of a $(x - 1)(x^2 + x + 1)$ when $x = \sqrt[3]{a + 1}$
- 2 c) Solve the simultaneous equations $xy = 8$ and $x + y = 6$
- 2 d) Simplify (i) $\frac{a - b}{b - a}$ (ii) $\frac{x - 3}{x^2 - 9}$
- 2 e) Simplify $\frac{x + 1}{1 - x} + \frac{x - 1}{1 + x}$
- 2 f) Find the value of x and y if $x + 2y + 2\sqrt{5} = 11 + \sqrt{x - y}$

- 3 Unit Only a) Solve $\frac{3}{x - 5} \leq 1$

- b) Write down a quadratic, in x , such that the quadratic is greater than or equal to zero if $x \leq -2$ or $x \geq 3$ and less than zero for any other values of x .

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Q1) a) $\frac{4+\sqrt{40}}{4}$

$= 2.58$

b) 2.65×10^9

c) $r = \sqrt{\frac{3V}{\pi h}}$

(i) $r = \sqrt{\frac{3 \times 1000}{\pi \times 50}}$

$= 4.4 \text{ cm}$

(ii) $r = \sqrt{\frac{3000}{\pi}}$

$r^2 = \frac{3000}{\pi}$

$r^3 = \frac{300}{\pi}$

$r = 9.4 \text{ cm.}$

d) $-12 \leq \frac{x}{3} \leq 4$

$-36 \leq 2x \leq 12$

$-18 \leq x \leq 6$



e) False: swimmers not all of the same ability.

(f) $\frac{6}{7-2\sqrt{5}}$

$= \frac{6}{7-2\sqrt{5}} \times \frac{7+2\sqrt{5}}{7+2\sqrt{5}}$

$= \frac{42+12\sqrt{5}}{49-20}$

$= \frac{42+12\sqrt{5}}{29}$

Ques 2

a) $|1-2x| = 5$

$1-2x = 5$ or $1-2x = -5$

$-2x = 4$ $-2x = -6$

$x = -2$ $x = 3$

b) $(x-1)^2 = 7$

$x-1 = \sqrt{7}$ or $x-1 = -\sqrt{7}$

$x = 1+\sqrt{7}$ or $1-\sqrt{7}$

c) $60\% = \$156$

$10\% = 2.6$

$100\% = \$260$

$\therefore$ original price = \$260

d) $x\sqrt{3} + 4\sqrt{y} = \sqrt{95} + \sqrt{40} + \sqrt{16}$

$= \sqrt{25} \cdot \sqrt{3} + \sqrt{16} \cdot \sqrt{5} + \sqrt{4} \cdot \sqrt{3}$

$= 5\sqrt{3} + 4\sqrt{5} + 2\sqrt{3}$

$= 7\sqrt{3} + 4\sqrt{5}$

$\therefore x = 7, y = 5$

e) let $x = 0.474747... (1)$

$\therefore 100x = 47.4747... (2)$

(2)-(1) $99x = 47$

$x = \frac{47}{99}$

$\therefore$ rational

(f) $|3x-2| \leq 4$

$-4 \leq 3x-2 \leq 4$

$-2 \leq 3x \leq 6$

$-\frac{2}{3} \leq x \leq 2$

Question 3

a) $x^3y - xy^3$

$= xy(x^2 - y^2)$

$= xy(x+y)(x-y)$

b) $\frac{1}{2} = \frac{1-2x}{2x}$

$= \frac{2-(1-2x)}{2x}$

$= \frac{1+2x}{2x}$

c) (i) $4-x = 5$

$-x = 1$

$x = -1$

(ii) $2x^2 - 7x - 4 = 0$

$(2x+1)(x-4) = 0$

$x = -\frac{1}{2}, 4$

(iii) $2-(1-x) \leq 3+2x$

$2-1+x \leq 3+2x$

$1+x \leq 3+2x$

$-2 \leq x$

(iv) $x^2 = 9x$

$x^2 - 9x = 0$

$x(x-9) = 0$

$x = 0, 9$

(d) $2^x = 2^{400} \div 2$

$2^x = 2^{399}$

$\therefore x = 399$

Ques 4

a) $|5-2x| \geq 3$

$5-2x \leq -3$ or $3 \leq 5-2x$

$-2x \leq -8$ or $-2 \leq -2x$

$x \geq 4$ $x \leq 1$

b) $\frac{3x-2}{4} - \frac{2x+1}{8} = 5$

$2(3x-2) - (2x+1) = 40$

$6x-4-2x-1 = 40$

$4x-5 = 40$

$4x = 45$

$x = 11\frac{1}{4}$

a) $a^2 - b^2 - 3a + 3b$

$= (a-b)(a+b) - 3(a-b)$

$= (a-b)(a+b-3)$

d) $8x^3 - 1000$

$= 8(x^3 - 125)$

$= 8(x-5)(x^2+5x+25)$

(e) $(x+\frac{1}{x})^2 = x^2 + 2 + \frac{1}{x^2}$

$\therefore x^2 + \frac{1}{x^2} = (x+\frac{1}{x})^2 - 2$

$= k^2 - 2$

(f) $3x-2y = 8 \dots (1)$

$5x-6y = 16 \dots (2)$

(1) $\times 3$ $9x-6y = 24 \dots (3)$

(3)-(2) $4x = 8$

$x = 2$

Subst $x=2$ into (1)

$6-2y = 8$

$y = -1$

$\therefore x = 2, y = -1$

Ques 5

a) $|x+7| = 3x-1$

$x+7 = 3x-1$ or $x+7 = -(3x-1)$

$8 = 2x$

$4 = x$

$4x = -6$

$x = -1\frac{1}{2}$

Test solutions:

$x = 4$

$x = -1\frac{1}{2}$

$|4+7| = 12-1$

True.

$|-1\frac{1}{2}+7| = -4-5-1$

False

$\therefore x = 4$

(b) $(x-1)(x^2+x+1)$

$= x^3 - 1$

$= (3\sqrt{9+1})^3 - 1$

$= 9+1-1$

$= 9$

(c) $xy = 8 \dots (1)$

$2x+y = 6 \dots (2)$

from (2) $y = 6-2x$

Subst into (1)

$x(6-2x) = 8$

$6x-x^2 = 8$

$x^2-6x+8 = 0$

$(x-4)(x-2) = 0$

$x = 2, \text{ or } 4$

Subst into (2)

$y = 4 \text{ or } 2$

$\therefore x = 2 \text{ or } x = 4$
 $y = 4 \text{ or } y = 2$

d) (i) $\frac{a-b}{b-a}$

$= -1$

(ii) $\frac{x-3}{x^2-9}$

$= \frac{x-3}{(x+3)(x-3)}$

$= \frac{1}{x+3}$

e) $\frac{x+1}{1-x} + \frac{x-1}{1+x}$

$= \frac{(1+x)^2 + (x-1)(1-x)}{(1-x)(1+x)}$

$= \frac{1+2x+1+x^2-1+x}{(1-x)(1+x)}$

$= \frac{4x+2}{(1-x)(1+x)}$

(f) $x+2y+2\sqrt{5} = 11+\sqrt{x-y}$

$x+2y+\sqrt{20} = 11+\sqrt{x-y}$

$\therefore x+2y = 11 \dots (1)$

$x-y = 20 \dots (2)$

(1)-(2) $3y = -9$

$y = -3$

Subst. $y = -3$ into (2)

$x+3 = 20$

$\therefore x = 17$

$\therefore x = 17, y = -3$

3 UNIT

a) $\frac{3}{x-5} \leq 1$

critical pts: $x = 5$

and $\frac{3}{x-5} = 1$

$3 = x-5$

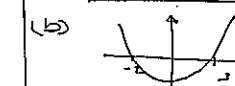
$8 = x$

test $x = 0$

$\frac{3}{-5} \leq 1$

true

$\therefore x < 5 \text{ or } 8 \leq x$



$y = (x+2)(x-3)$

$y = x^2 - x - 6$